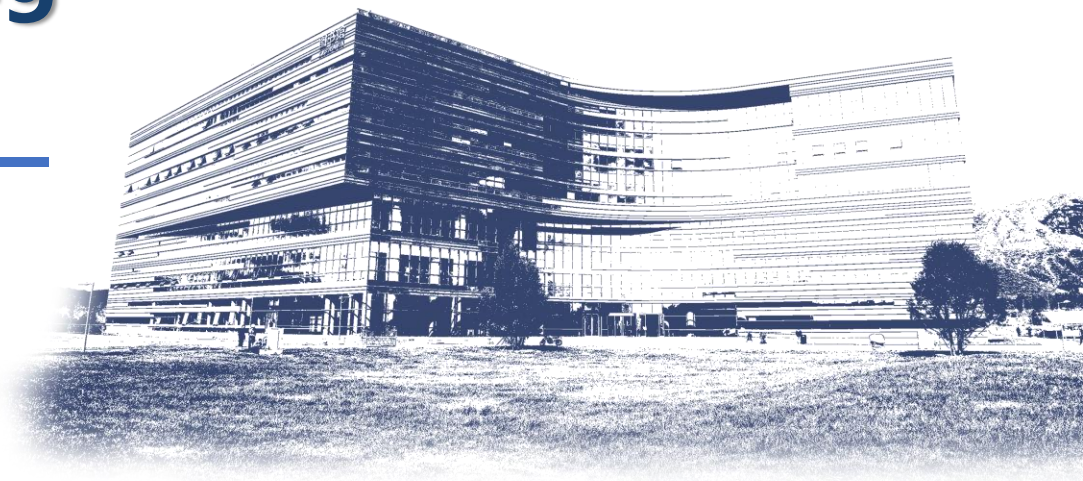
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2.Understanding and Modeling Organizational Systems

System Analysis and Design



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Learning Objectives

- Understand that organizations and their members are systems and that analysts need to take a systems perspective.
- Depict systems graphically using context-level data flow diagrams, and entity-relationship models, use cases, and use case scenarios.
- Recognize that different levels of management require different systems.
- Comprehend that organizational culture impacts the design of information systems.

Three Main Forces Interacting to Shape Organizations

- Levels of management
- Design of organizations
- Organizational cultures

Organizations are Composed of Interrelated Subsystems

- Influenced by levels of management decision makers that cut horizontally across the organizational system
  - Operations（业务层）
  - Middle management（中层管理者）
  - Strategic management（决策管理者）
- Influenced by organizational cultures and subcultures

Major Topics

- Organizations as systems
- Depicting systems graphically
  - Data flow diagram
  - Entity-relationship model
  - Use case modeling
- Levels of management
- Organizational culture

2.1 Organizations as Systems

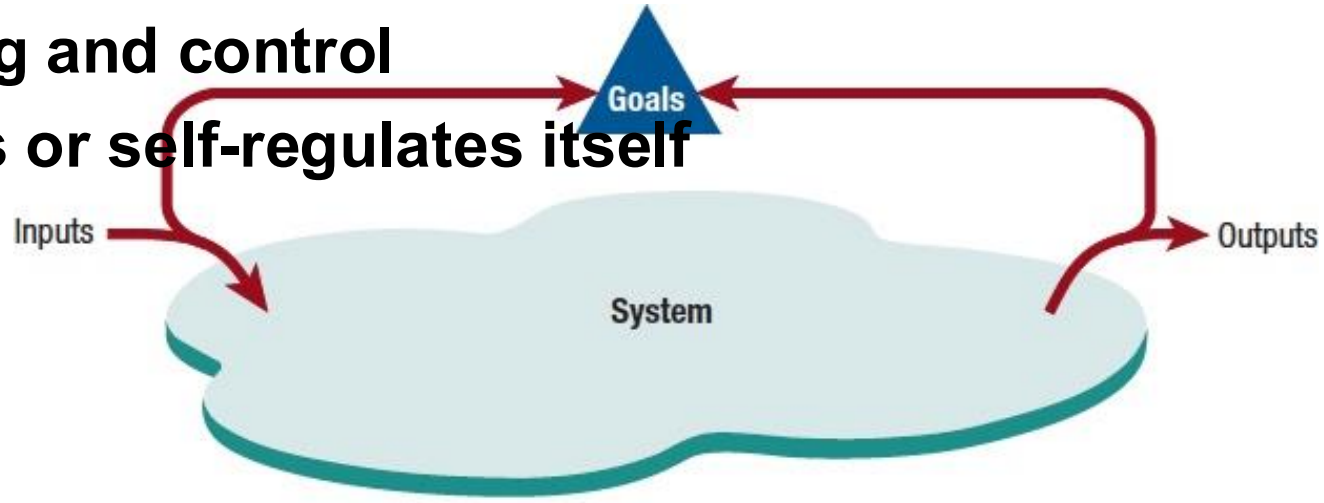
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- Conceptualized as systems designed to accomplish predetermined goals and objectives
- Composed of smaller, interrelated systems serving specialized functions
- Specialized functions are reintegrated to form an effective organizational whole

2.1.1 Interrelatedness and Independence of Systems

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- All systems and subsystems are interrelated and interdependent
- All systems process inputs from their environments
- All systems are contained by boundaries separating them from their environments
- System feedback for planning and control
- An ideal system self-corrects or self-regulates itself



System Outputs Serve as Feedback That Compares Performance with Goals

Organizational Environments

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- Community
  - Physical location
  - Demographic profile (education, income)
- Economic
  - Market factors
  - Competition
- Political
  - State and local government
- Legal
  - Federal, state, regional, local laws, and guidelines

2.1.2 Virtual Organizations and Virtual Teams

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- A virtual organization has parts of the organization in different physical locations
- Computer networks and communications technology are used to bring virtual teams together to work on projects

Benefits of Virtual Organizations and Teams

- Possibility of reducing costs of physical facilities
- More rapid response to customer needs
- Helping virtual employees to fulfill their familial obligations to children or aging parents

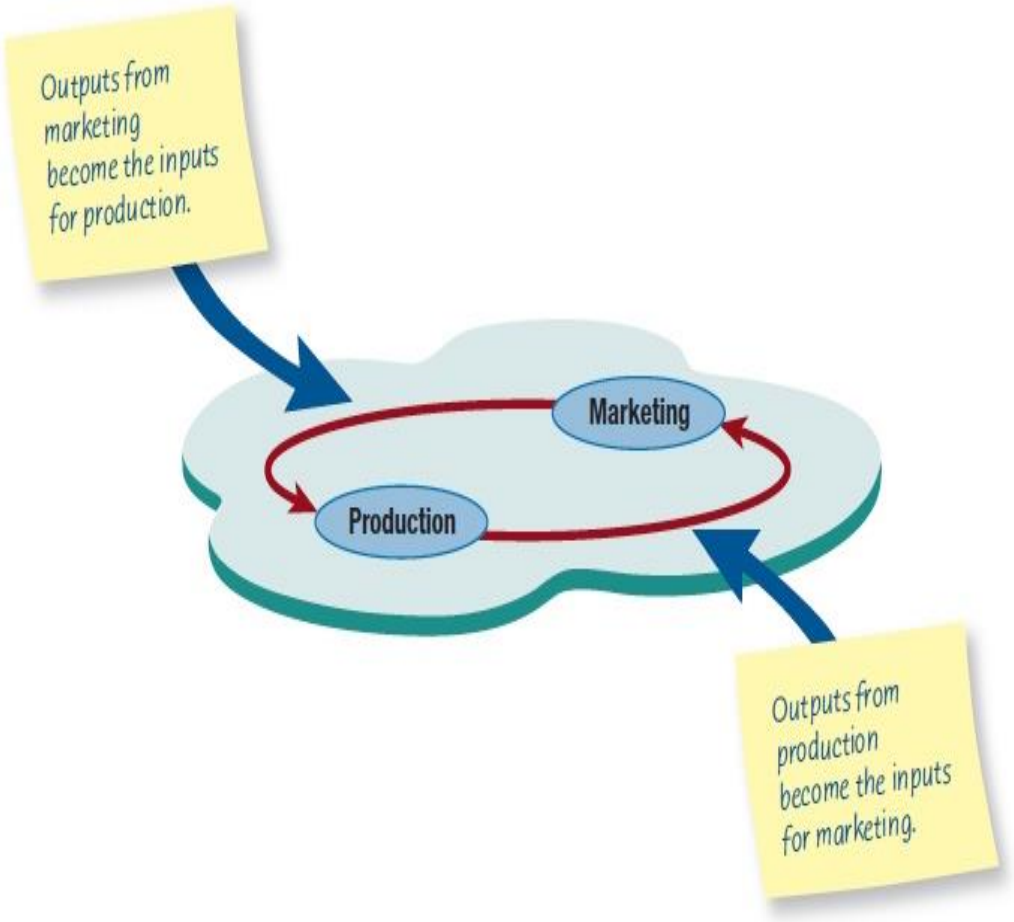
2.1.3 Taking a Systems Perspective

- Allows system analyst to understand businesses they will come into contact with
- It is important that members of subsystems realize that they are interrelated with other subsystems
- Problems occur when each manager thinks that his/her department is the most important
- Bigger problems may occur when that manager rises through the ranks

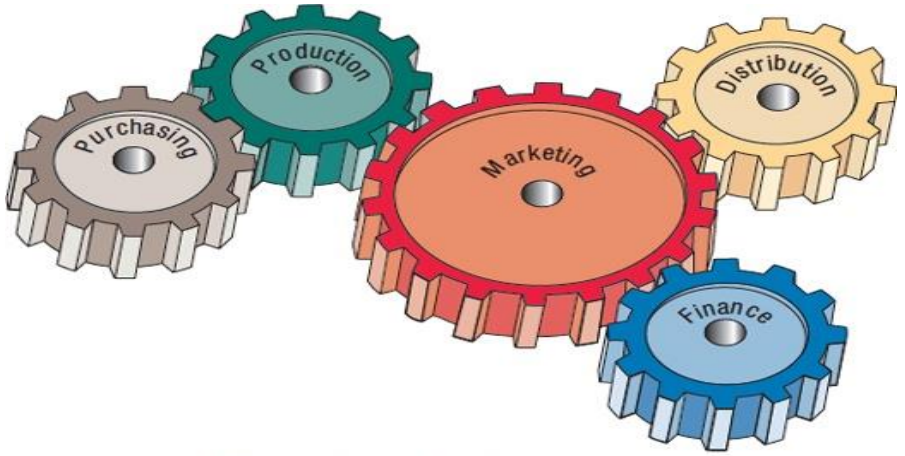
Systems Perspective

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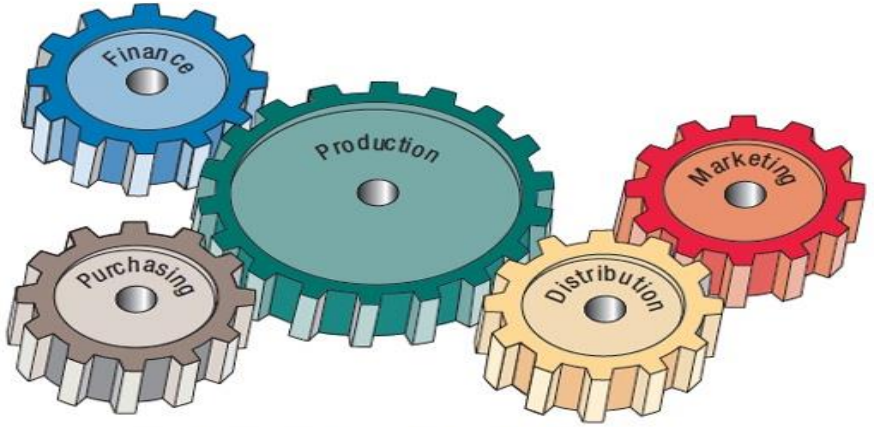
Taking a Systems Perspective



Perspective of Functional Managers



How a Marketing Manager May View the Organization



How a Production Manager May See the Organization

2.1.4 Enterprise Resource Planning

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- Enterprise Resource Planning (ERP) is an integrated organizational information system
- Software that helps the flow of information between the functional areas within the organization
- Recently ERP systems are moving to cloud computing

ERP and the Organization

- ERP can affect every aspect of the organization, including:
  - Design of employees' work
  - Skills required for job competency
  - Strategic positioning of the company

Issues to Be Overcome for ERP Success

- Many issues must be overcome for the ERP installation is to be declared a success:
  - User acceptance
  - Integration with legacy systems and the supply chain
  - Upgrading functionality (and complexity) of ERP modules
  - Reorganizing work life of users and decision makers
  - Expanded reach across several organizations
  - Strategic repositioning of the company

2.2 Depicting Systems Graphically

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- Context-level data flow diagrams
- Entity-relationship model
- Use case modeling

2.2.1 Context-Level Data Flow Diagrams

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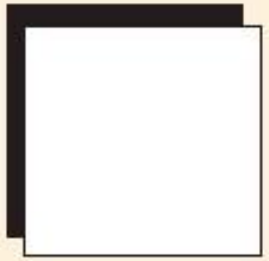
- Focus is on the data flowing into and out of the system and the processing of the data
- Shows the scope of the system:
  - What is to be included in the system
  - The external entities are outside the scope of the system

The Basic Symbols of a Data Flow Diagram

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A **process** means that some action or group of actions take place.



An **entity** is a person, group, department, or any system that either receives or originates information or data.

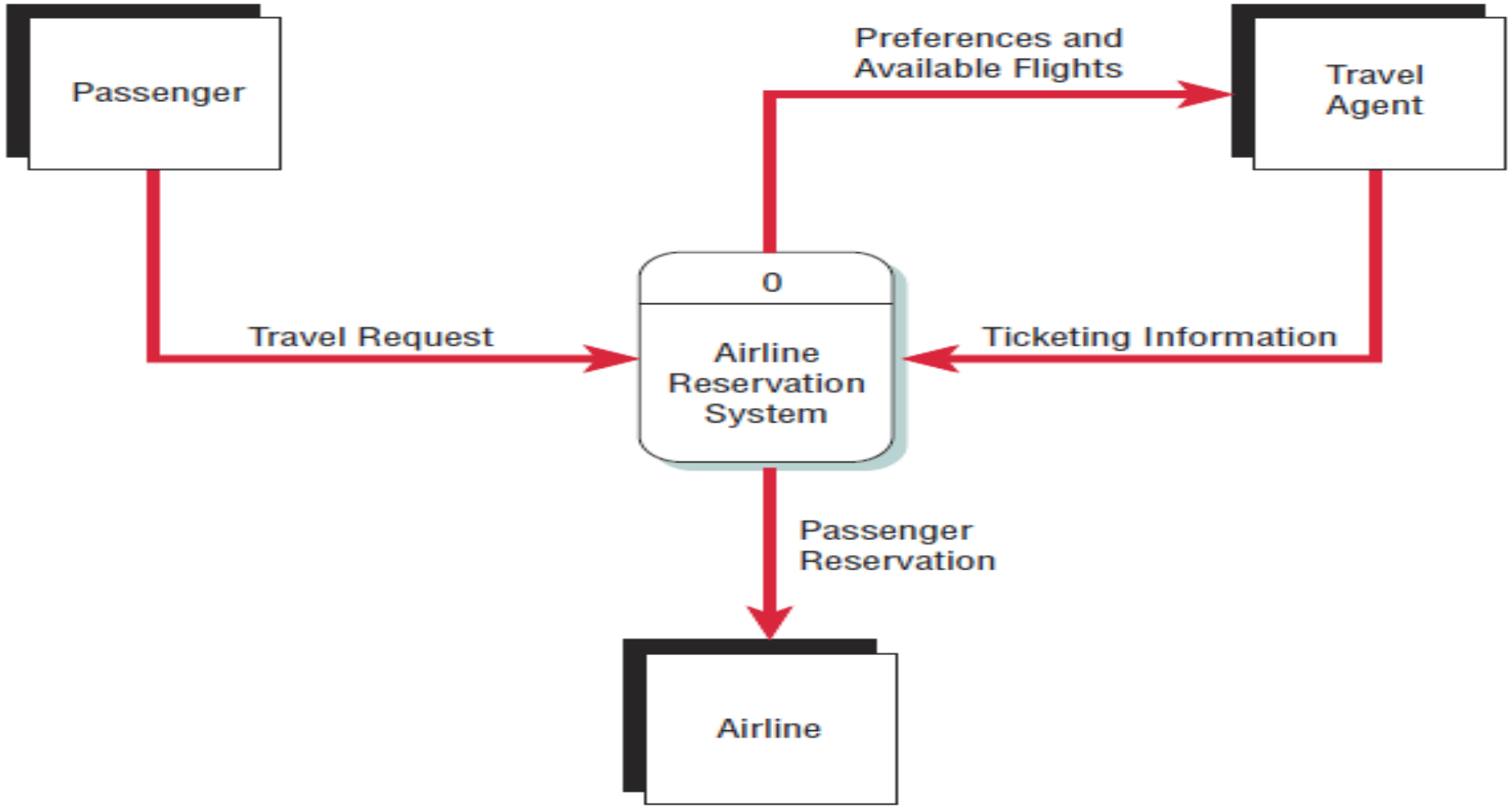


A **data flow** shows that information is being passed from or to a process.

Airline Reservation System

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A context-level data flow diagram for an airline reservation system



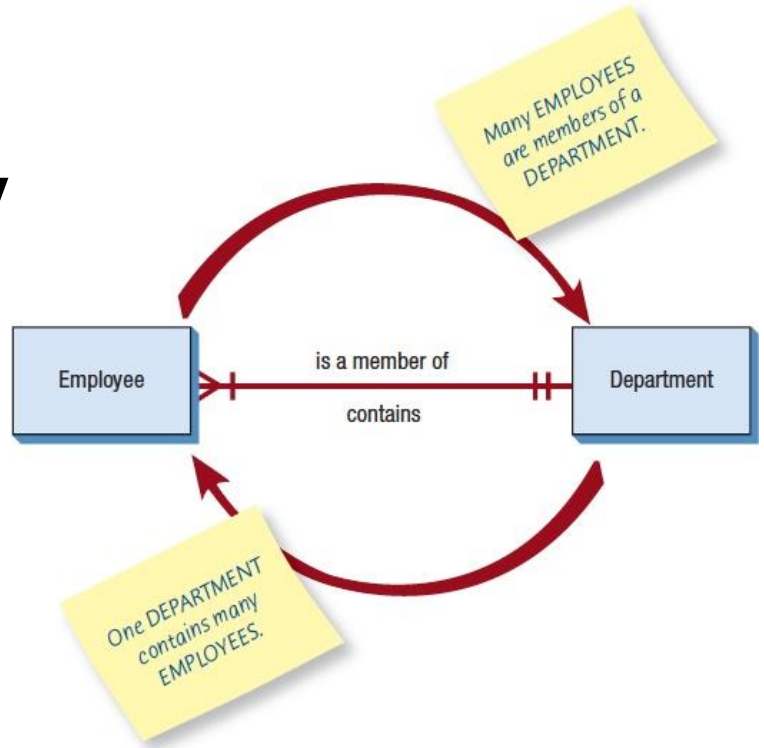
2.2.2 Entity-Relationship Model

- **Focus is on the entities and their relationships within the organizational system**
- **Another way to show the scope of a system**

Relationships

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- Relationships show how the entities are connected
- Three types of relationships:
  - One-to-one
  - One-to-many
  - Many-to-many



An entity-relationship diagram showing a many-to-one relationship

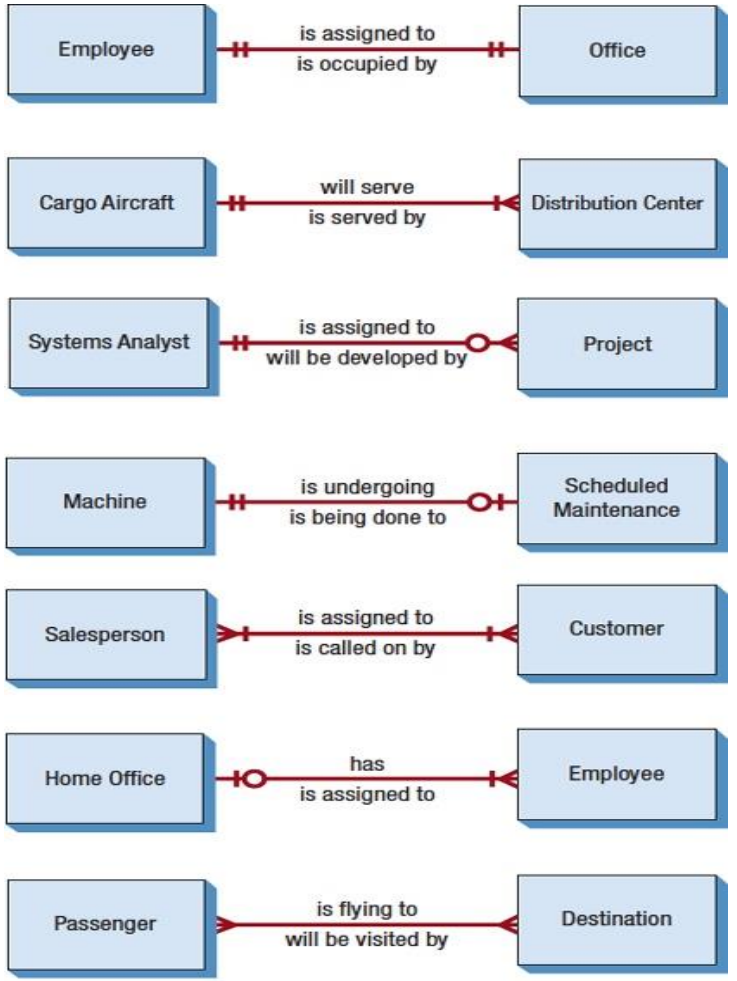


Figure. Examples of Different Types of Relationships in E-R Diagrams

Entities

- **Fundamental entity**
- **Associative entity**
- **Attributive entity**

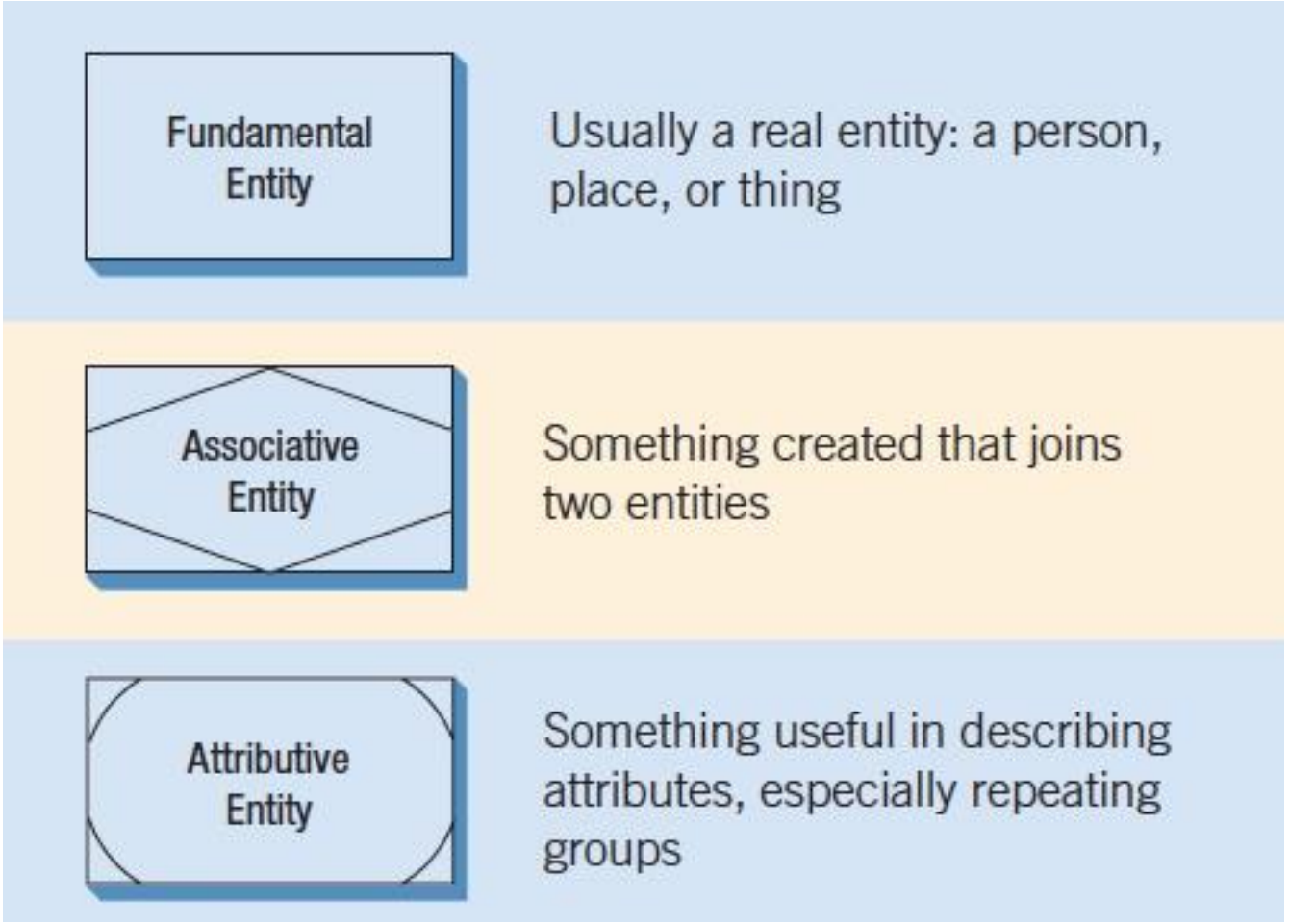
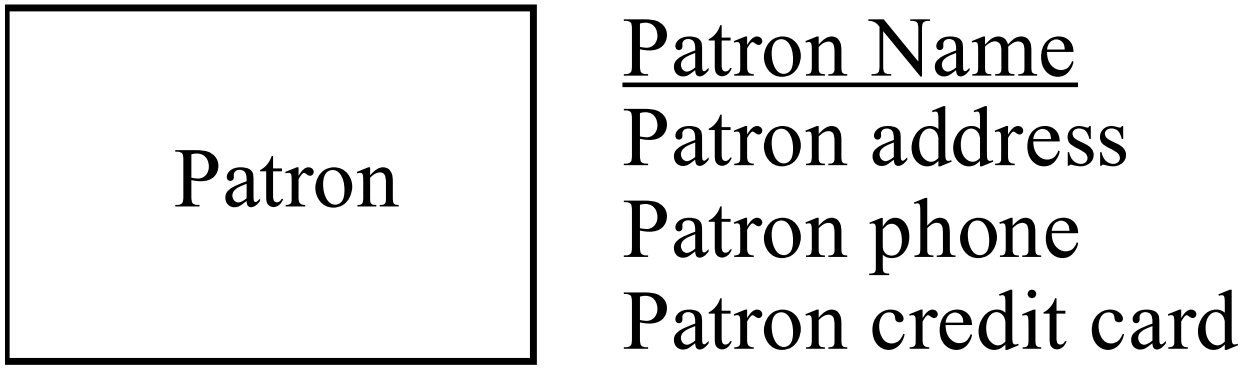


Figure. Three Different Types of Entities Used in E-R Diagrams

Attributes

- **Data attributes may be added to the diagram.**



Creating Entity-Relationship Diagrams

- List the entities in the organization
- Choose key entities to narrow the scope of the problem
- Identify what the primary entity should be
- Confirm the results of the above through data gathering

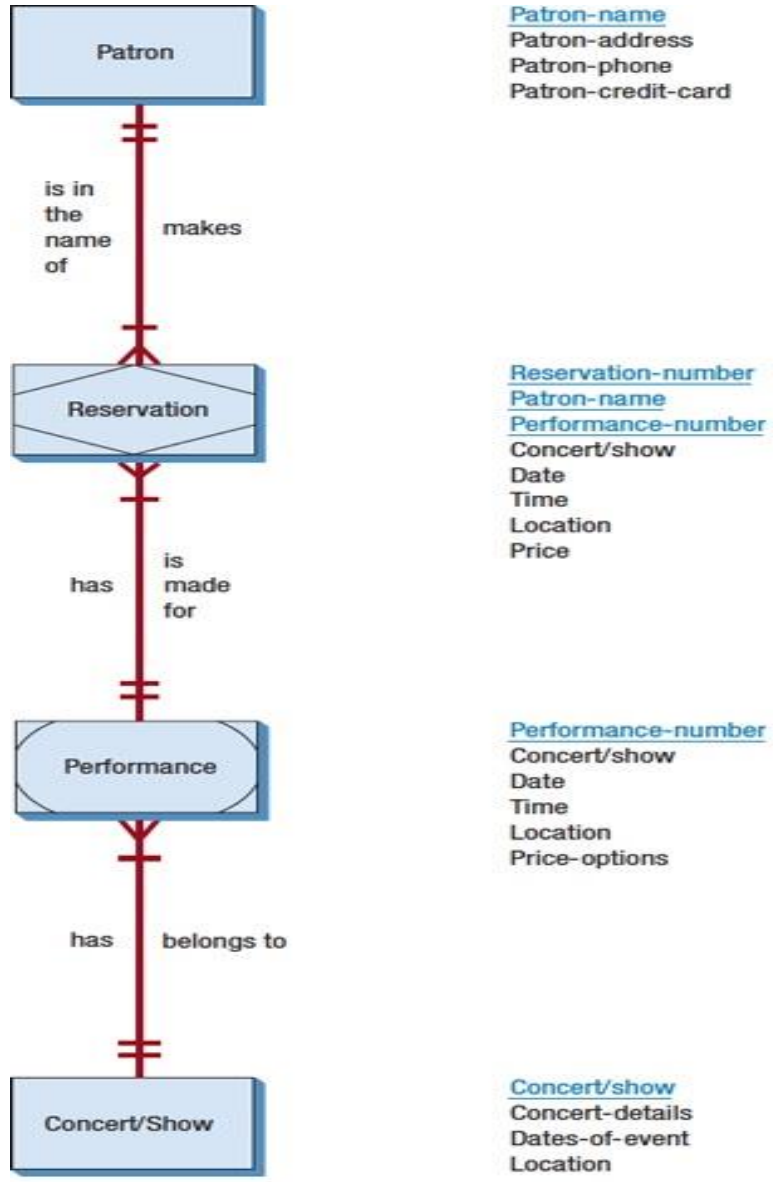


Figure. A More Complete E-R Diagram Showing Data Attributes of the Entities

2.3 Use Case Modeling

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- Part of the unified modeling language (UML)
- Describes what a system does without describing how the system works
- A view of the system requirements
- Analyst works with business experts to develop requirements

2.3.1 Use Case Diagram

- Actor
  - Refers to a particular role of a user of the system
  - Similar to external entities; they exist outside of the system
- Use case symbols
  - An oval indicating the task of the use case
- Connecting lines
  - Arrows and lines used to diagram behavioral relationships

Actor

- Divided into two groups
  - Primary actors:
    - Supply data or receive information from the system
    - Provide details on what the use case should do
  - Supporting actors:
    - Help to keep the system running or provide help
    - The people who run the help desk, the analysts, programmers, and so on

A Use Case Always Provides Three Things

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- An actor that initiates an event
- The event that triggers a use case
- The use case that performs the actions triggered by the event

2.3.2 Use Case Relations

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- Behavioral relationships（行为关系）
  - Communicates（交流）
    - Used to connect an actor to a use case
  - Includes（包含）
    - Describes the situation in which a use case contains behavior that is common to more than one use case
  - Extends（扩展）
    - Describes the situation in which one use case possesses the behavior that allows the new case to handle a variation or exception from the basic use case
  - Generalizes（泛化）
    - Implies that one thing is more typical than the other thing

Four Types of Behavioral Relationships and the Lines Used to Diagram Each

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Relationship	Symbol	Meaning
Communicates	———	An actor is connected to a use case using a line with no arrowheads.
Includes	<<include>> ◄- - - -	A use case contains a behavior that is common to more than one other use case. The arrow points to the common use case.
Extends	<<extend>> - - - - ►	A different use case handles exceptions from the basic use case. The arrow points from the extended to the basic use case.
Generalizes	——▷	One UML "thing" is more general than another "thing." The arrow points to the general "thing."

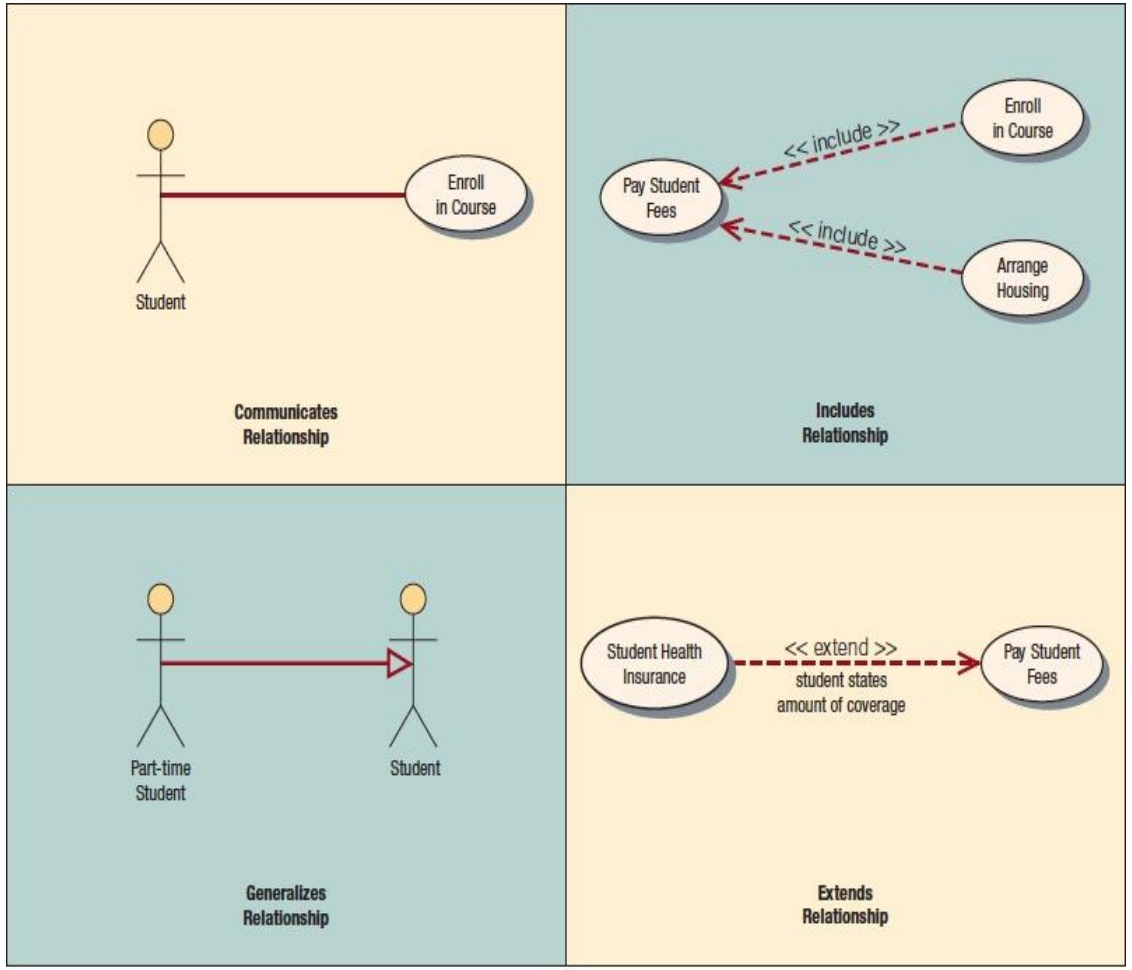


Figure. Actors, Use Cases, and Relationships for a Student Enrollment Example

2.3.3 Scope（开发系统的范围）

- System scope defines its boundaries:
  - What is in or outside the system
  - Project has a budget that helps to define scope
  - Project has a start and an end time
- Actors are always outside of scope
- Communication lines are the boundaries and define the scope

2.3.4 Developing Use Case Diagrams

- Review the business specifications and identify the actors involved
- Identify the high-level events and develop the primary use cases that describe those events and how the actors initiate them
- Review each primary use case to determine the possible variations of flow through the use case
- The context-level data flow diagram could act as a starting point for creating a use case

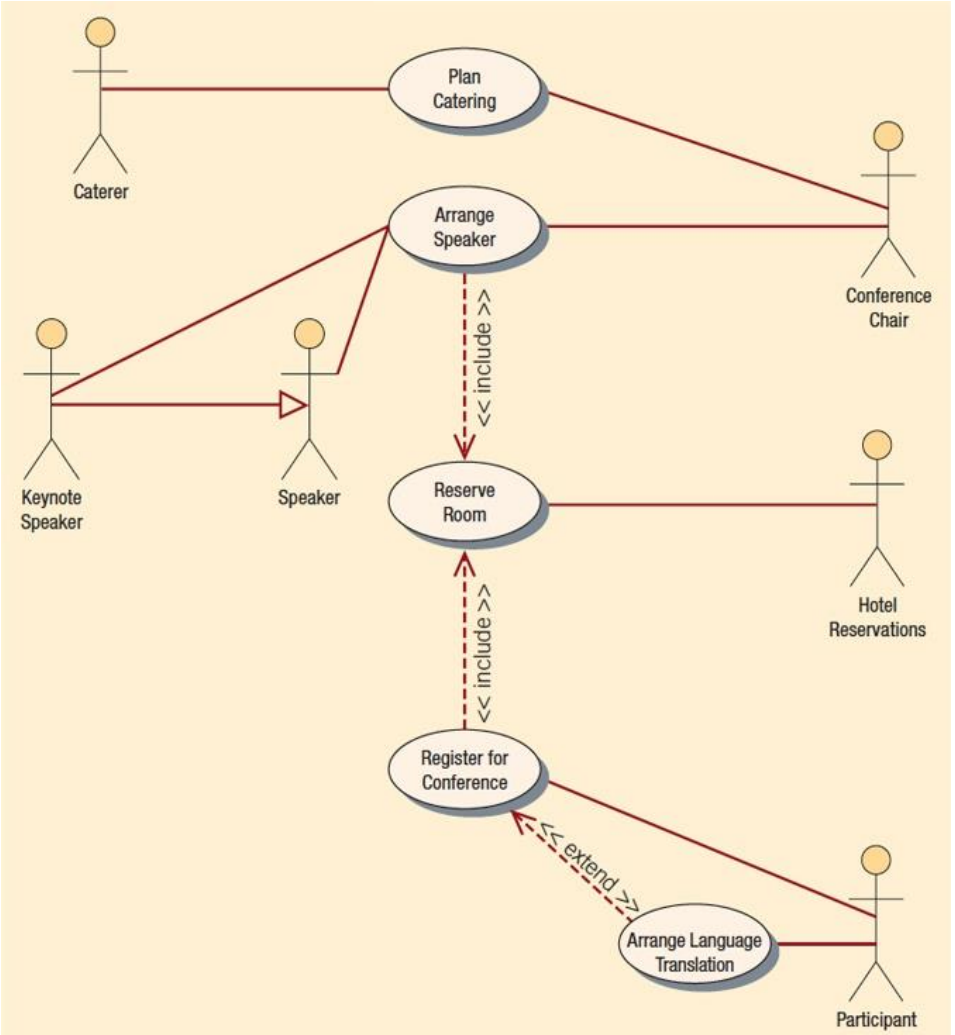
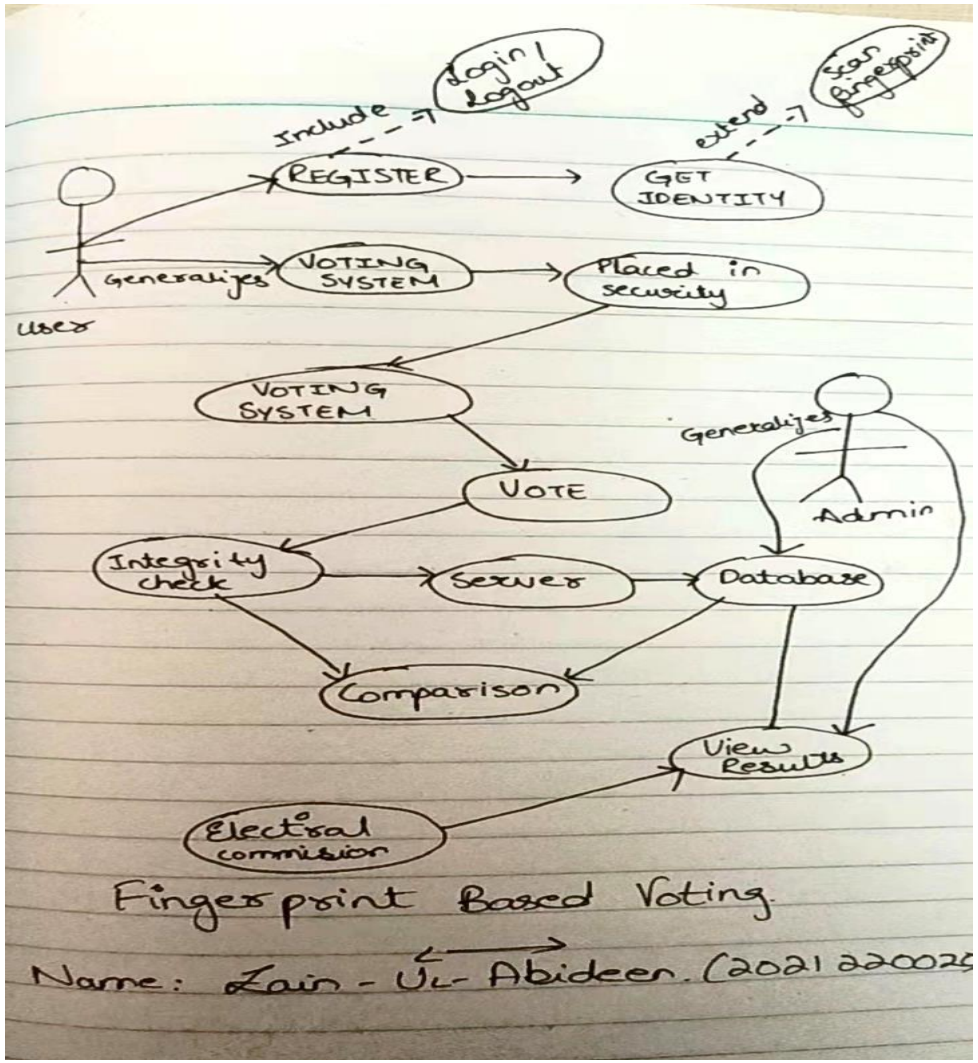
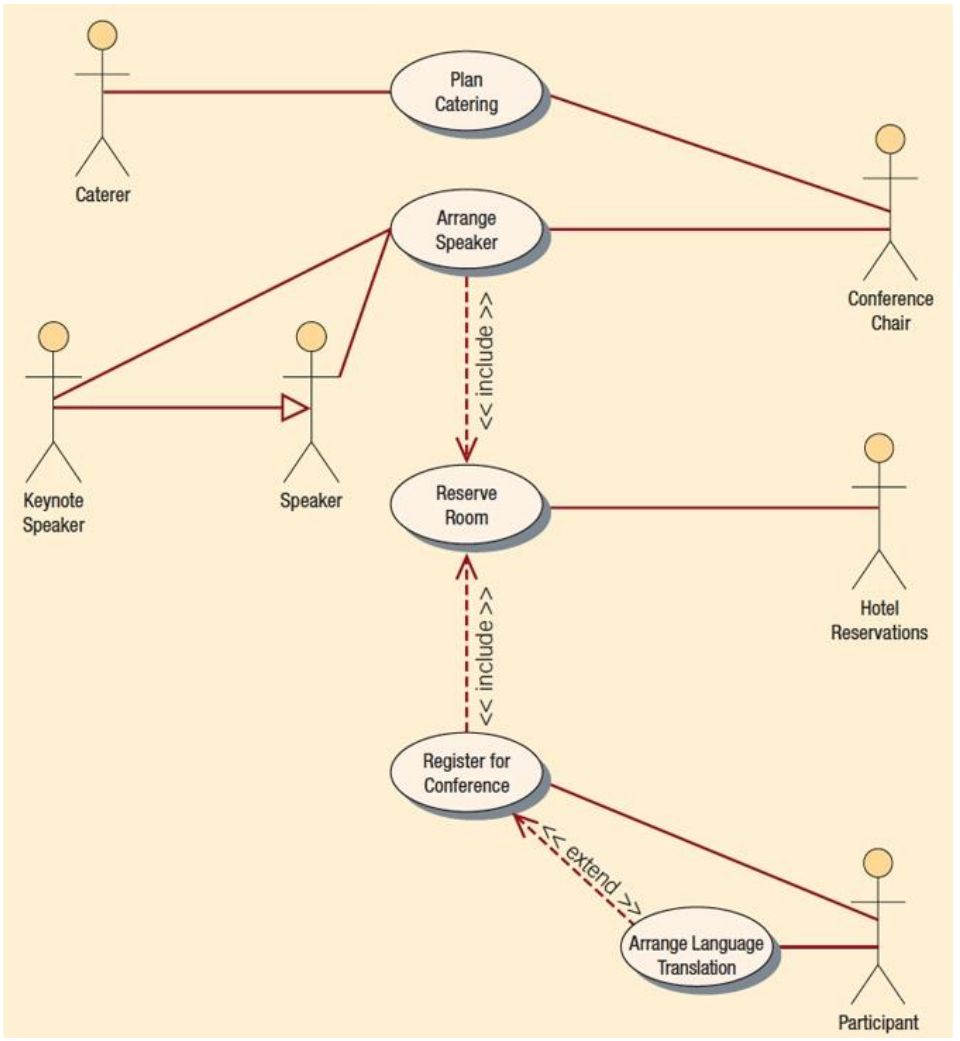


Figure. A Use Case Diagram Representing a System Used to Plan a Conference


Login / Logout
Include
Scan fingerprint
extend
REGISTER
GET IDENTITY
Generalizes
VOTING SYSTEM
Placed in security
User
VOTING SYSTEM
Generalizes
VOTE
Admin
Integrity check
Server
Database
Comparison
View Results
Electoral commission
Fingerprint Based Voting
Name: Zain - Ul- Abideen. (2021220025


Caterer
Plan Catering
Arrange Speaker
Conference Chair
<< include >>
Keynote Speaker
Speaker
Reserve Room
Hotel Reservations
<< include >>
Register for Conference
<< extend >>
Arrange Language Translation
Participant

2.3.5 Developing the Use Case Scenarios

开发用例场景

- The description of the use case
- Three main areas:
  - Use case identifiers and initiators
  - Steps performed
  - Conditions, assumptions, and questions

Use case name:	Register for Conference	UniqueID: Conf RG 003
Area:	Conference Planning	
Actor(s):	Participant	
Stakeholder:	Conference Sponsor, Conference Speakers	
Level:	Blue	
Description:	Allow conference participant to register online for the conference using a secure website.	
Triggering Event:	Participant uses Conference Registration website, enters userID and password, and clicks the logon button.	
Trigger type:	External Temporal	
Steps Performed (Main Path)		**Information for Steps**
1. Participant logs in using the secure web server.		userID, Password
2. Participant record is read and password is verified.		Participant Record, userID, Password
3. Participant and session information is displayed on the Registration web page.		Participant Record, Session Record
4. Participant enters information on the Registration Web form and clicks **Submit** button.		Registration Web Form
5. Registration information is validated on the Web server.		Registration Web Form
6. Registration Confirmation page is displayed to confirm registration information.		Confirmation Web Page
7. Credit card is charged for registration fees.		Secure Credit Card Web Page
8. Add Registration Journal record is written.		Confirmation Web Page
9. Registration record is updated on the Registration Master.		Confirmation Web Page, Registration Record
10. Session record is updated for each selected session on the Session Master.		Confirmation Web Page, Session Record
11. Participant record is updated for the participant on the Participant Master.		Confirmation Web Page, Participant Record
12. Successful Registration Confirmation web page is sent to the participant.		Registration Record Confirmation Number
Preconditions:	Participant has already registered and has created a user account.	
Postconditions:	Participant has successfully registered for the conference.	
Assumptions:	Participant has a browser and a valid userID and password.	
Success Guarantee:	Participant has registered for the conference and is enrolled in all selected sessions.	
Minimum Guarantee:	Participant was able to logon.	
Requirements Met:	Allow conference participants to be able to register for the conference using a secure website.	
Outstanding Issues:	How should a rejected credit card be handled?	
Priority:	High	
Risk:	Medium	

Figure. A Use Case Scenario is Divided into Three Sections

Use Case Header Area

- Has a name and a unique ID
- Include application area
- List actors
- Include stakeholders
- Include the level
- Has a brief description of the use case

2.3.6 Use Case Levels

- Use case levels describe how global or detailed the use case description is:
  - White (like clouds): enterprise level
  - Kite: business unit or department level
  - Blue (sea level): user goals
  - Indigo (or fish): functional or subfunctional
  - Black (or clam): most detailed

Alternative Scenarios （可能情况）

- Extensions or exceptions to the main use case
- Number with an integer, decimal point, integer
- Steps that may or may not always be used

Use Case Footer Area

- Preconditions—need to be met before use case can be performed
- Postconditions or the state of the system after the use case has finished
- Assumptions
- Minimal guarantee
- Success guarantee
- Outstanding issues
- Optional priority and risk

2.3.7 Four Steps Used to Create Use Cases

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- Use agile stories, problem definition objectives, user requirements, or a features list
- Ask about the tasks that must be done
- Determine if there are any iterative or looping actions
- The use case ends when the customer goal is complete

2.3.8 Why Use Case Diagrams Are Helpful

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- Identify all the actors in the problem domain
- Actions that need to be completed are also clearly shown on the use case diagram
- The use case scenario is also worthwhile
- Simplicity and lack of technical detail

The Main Reasons for Writing Use Cases are Their Effectiveness in Communicating with Users and Their Capturing of User Stories

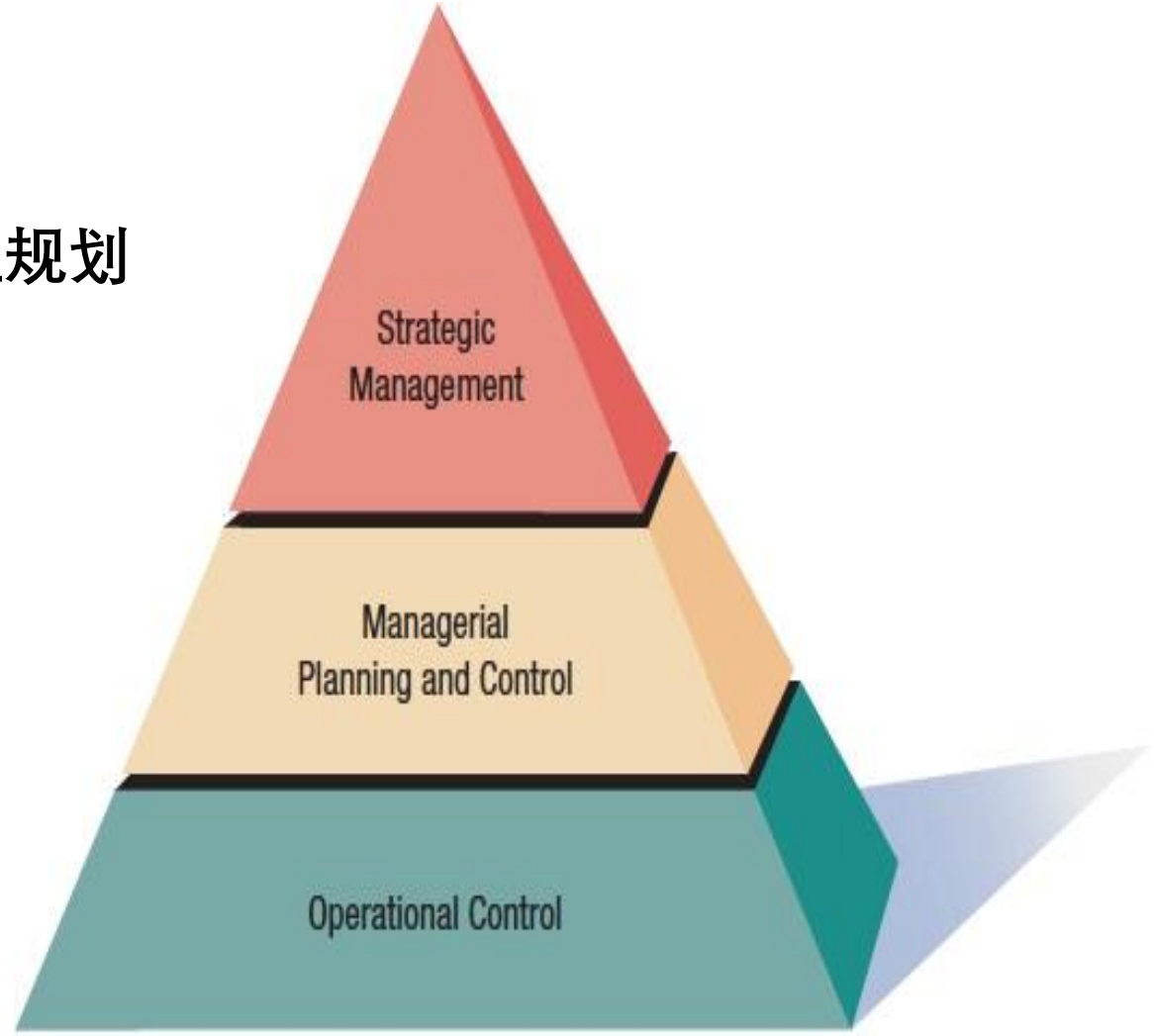
- Use cases effectively communicate systems requirements because the diagrams are kept simple.
- Use cases allow people to tell stories.
- Use case stories make sense to nontechnical people.
- Use cases do not depend on a special language.
- Use cases can describe most functional requirements (such as interactions between actors and applications).
- Use cases can describe nonfunctional requirements (such as performance and maintainability) through the use of stereotypes.
- Use cases help analysts define boundaries.
- Use cases can be traceable, allowing analysts to identify links between use cases and other design and documentation tools.

2.4 Management in Organizations Exists on Three Horizontal Levels

Operational control（运营控制），

managerial planning and control（管理规划和控制），

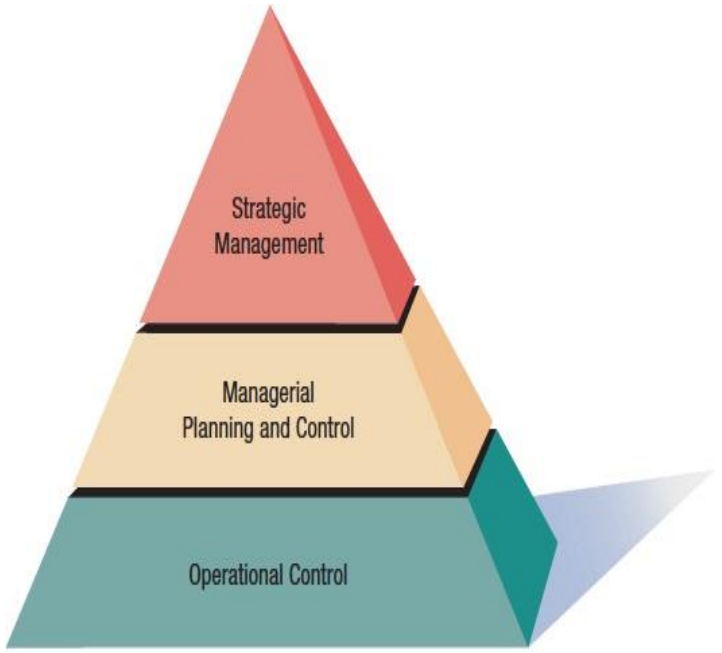
strategic management（战略管理）.



Operations Control

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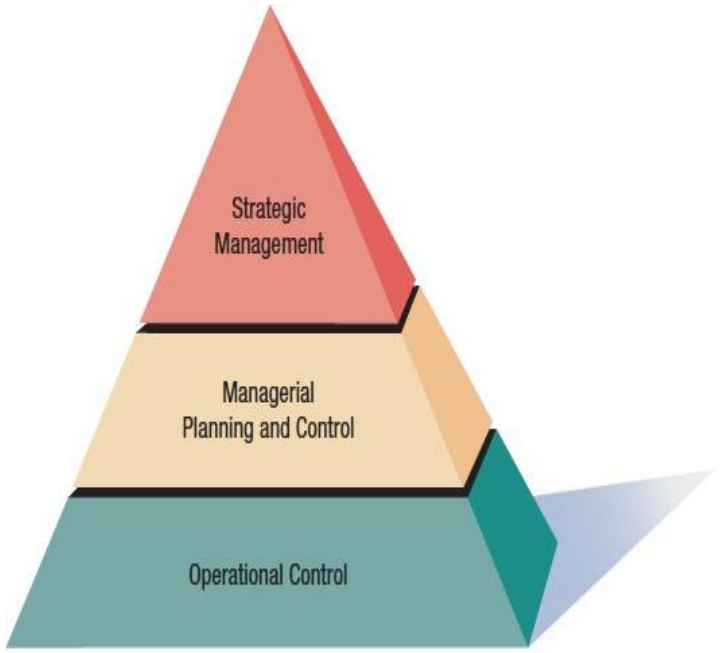
- Make decisions using predetermined rules that have predictable outcomes
- Oversee the operating details of the organization



Managerial Planning and Control

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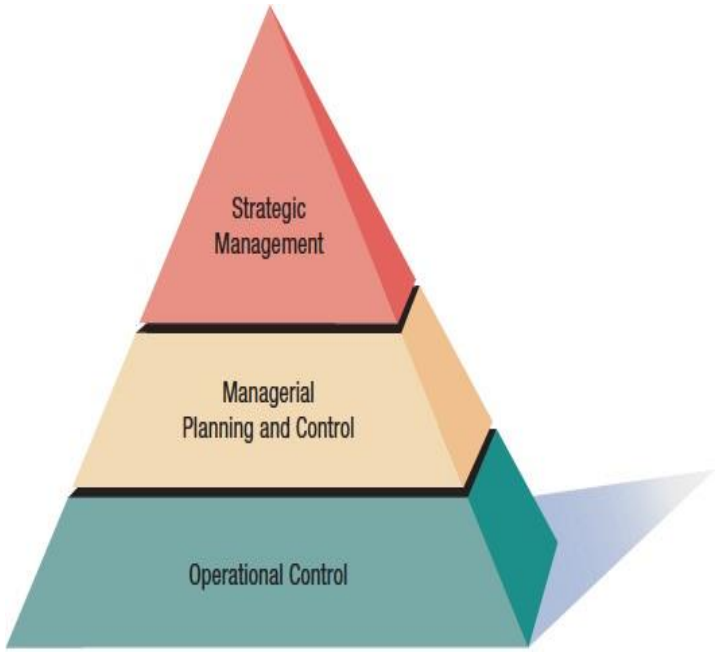
- **Make short-term planning and control decisions about resources and organizational objectives**
- **Decisions may be partly operational and partly strategic**



Strategic Management

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- Look outward from the organization to the future
- Make decisions that will guide middle and operations managers
- Work in highly uncertain decision-making environment
- Faced with semistructured problems
- Define the organization as a whole



2.4.1 Managerial Levels（管理水平）

- Different organization structure
- Leadership style
- Technological considerations
- Organization culture
- Human interaction
- All carry implications for the analysis and design of information systems

2.4.2 Collaborative Design（系统设计）

- External and internal stakeholders follow processes to share in designing a system to meet their goals
- Giving power to those who possess a technical or strategic expertise

2.5 Organizational Culture

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- Organizations have cultures and subcultures
- Learn from verbal and nonverbal symbolism

Technology Impact on Culture

- Technology is changing the culture of organizations and teams
- Slack is an employer-sanctioned social media platform, or workplace-messaging app
- Public and private channels
- Direct or group messages

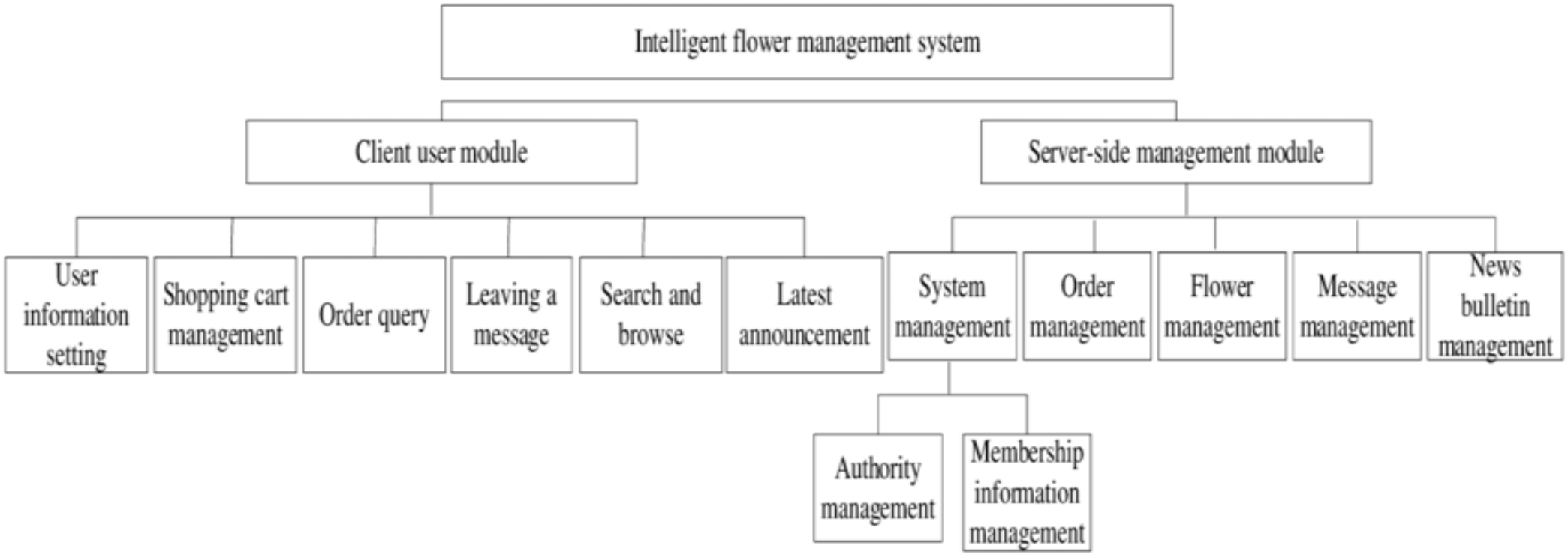
Summary

- Organizational fundamentals
  - Organizations as systems
  - Levels of management
  - Organizational culture
- Graphical representation of systems
  - DFD
  - ERD
  - Use case diagrams and scenarios
- Levels of managerial control
  - Operational
  - Middle management
  - Strategic
- Organizational culture

Homework of Week 2

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- Requirement (text)
- Functional structure of your system (figure)



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Thank for your Listening.

System Analysis and Design

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